

RESEARCH ARTICLE

The effect of moisture during development on phenotypes of egg-laying reptiles: a systematic review and meta-analysis

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ABSTRACT

The embryonic environment is critical for the development of many ectothermic vertebrates, which makes them highly vulnerable to environmental change. Changes in temperature and moisture, in particular, are known to influence embryo survival and offspring phenotypes. Although most papers concerning phenotypic development of terrestrial ectotherms focus on the role of temperature on eggs and embryos, the comparatively small number of studies on the effects of substrate moisture are well suited for quantitative analysis aimed at guiding future research. To accomplish this goal, we compiled data from 37 studies on 28 different reptile species and used a meta-analytic approach to quantify the effect of substrate moisture on several offspring outcomes: hatching success (survival), incubation duration, hatchling mass and length, and sex ratio. We found that substrate moisture had a small effect across most traits but significantly affected body size (i.e. length and mass), with wetter conditions producing longer and heavier hatchlings. Temperature also moderated the effect of moisture on hatching success, with higher temperatures resulting in lower success. Additionally, the effect of moisture on hatching success and hatchling mass was enhanced by larger differences in moisture concentration between treatments, yet the effect was small. Lastly, substrate moisture affected sex ratio in turtles, but not in other squamates. Overall, these analyses provide a foundation for further research investigating the effects of moisture on oviparous reptile development. Increasing the diversity of environmental variables for which we understand their impact on animal phenotype will be beneficial in an era with wide-ranging global change.

KEY WORDS: Developmental plasticity, Ectotherm, Incubation, Squamate, Testudines, Water content

INTRODUCTION

Environmental factors experienced by embryos such as temperature, photoperiod, nutrition and moisture are known to shape an individual's phenotype in a process known as developmental plasticity (Monaghan, 2008; Uller, 2008; Sultan, 2017). For example, European map

butterflies (*Araschiria levana*) develop into different colour morphs when larvae are raised in cold spring temperatures versus in hot summer temperatures (Gilbert, 2001; Nijhout, 2003; Friberg and Karlsson, 2010). Developmental plasticity impacts how organisms respond to environmental change globally because phenotypic variance affects the opportunity for and strength of selection, and has consequences on ecological responses within and among species (Miner et al., 2005; Uller, 2008; Sultan, 2017; Noble et al., 2018a; Warner et al., 2018). Developmental plasticity can be adaptive, neutral or maladaptive depending on how the early developmental environment matches with the environment an individual will experience later in life (Monaghan, 2008; While et al., 2018). The potential for mismatched developmental and later life environments may present a challenge as rates of global environmental change increase. Thus, understanding the fitness consequences of environmentally driven developmental plasticity in a generalizable and comparative way will allow better anticipation of the impacts of global change.

Oviparous (egg-laying) reptiles experience a wide range of nest environments that are known to impact phenotype and survival (Mitchell et al., 2018; Refsnider et al., 2019), which makes them suitable organisms for examining the effects of environmental factors on phenotypic development. Eggs are influenced by the amount of water they absorb and the temperatures they endure via moisture and heat exchange between the egg, the air and the nest substrate (Ackerman, 1991; Kearney and Enriquez-Urzelai, 2023). These processes are guided by a strong interaction between temperature and moisture that, in concert, significantly impact embryo development (Ackerman, 1991; Rozen-Rechels et al., 2019; Kearney and Enriquez-Urzelai, 2023). Interestingly, nest temperature has been well studied and is well known to affect a range of traits in reptiles, including incubation duration, hatching success, sex ratio, morphology, growth rate, behavior, performance and physiology (Noble et al., 2018a; Refsnider et al., 2019; Raynal et al., 2022). The impacts of temperature on phenotype can affect fitness over short and long time periods (Noble et al., 2018a). Further, the effects of substrate moisture have been documented in empirical studies (Packard et al., 1980, 1983, 1987, 1988) and subsequent literature reviews from the 1990s (Ackerman, 1991; Packard, 1991; Packard and DeMarco, 1991; Vleck, 1991). Since then, there has been more research on this topic and incorporating this newer information about the effect of soil moisture on reptile phenotype will improve understanding of general trends. Additionally, our changing climate is predicted to produce increasingly dry substrate conditions, which will test our understanding of the interaction between temperature and soil moisture (IPCC, 2014; Vázquez et al., 2017; While et al., 2018; Doody et al., 2020). Overall, it is difficult to anticipate the potential impact of changes in climate on reptiles without a current and comprehensive understanding of how substrate moisture impacts their developmental plasticity.

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The exchange of moisture between reptile eggs and the surrounding substrate is influenced by substrate water potential. The substrate water potential is the combined interaction of osmotic pressure (often measured in kPa and reflects the amount of water in the substrate) and matrix potential, which refers to a weak molecular interaction between substrate surface particles and water molecules (Ackerman, 1991). Substrates that are 'dry' (large, negative water potential values) have less water available for eggs to absorb, which may lead to desiccation during development. In contrast, substrates that are fully saturated with water (water potential = 0 kPa) may prevent gas exchange across the eggshell, preventing successful development of embryos. Thus, soil moisture levels have great potential to impact oviparous reptile development.

Indeed, previous research suggests that substrate moisture can have a strong impact on reptile development (Packard and Packard, 1988; Ackerman, 1991; Packard, 1991). Wet nests with poor drainage tend to have lower hatching success than those with higher drainage, as poor drainage nests become inundated with water (Markle et al., 2021). However, wetter nests within tolerated hydric limits can lead to larger hatchlings (Gatto and Reina, 2022; Phillips et al., 1990; Warner and Andrews, 2002b). Eggshell characteristics may also be important, as studies show that species with flexible, soft-shelled eggs may more readily convert yolk to hatchling tissue while incubating compared with those with rigid eggshells, thereby producing larger hatchlings in wetter nests (Belinsky et al., 2004; Booth, 2004; Deeming and Unwin, 2004). Thus, it is likely that water uptake by reptile eggs affects embryo metabolism, offspring mass, survival and other components of fitness. Importantly, shell permeability may influence how water passes across the eggshell, and eggshell composition may also have some effect on water uptake (Tracy, 1980; Ackerman, 1991; Packard and DeMarco, 1991). Overall, strong evidence indicates that substrate moisture can impact the development of certain oviparous reptiles, but we do not fully understand the general effects of moisture across taxa.

The effect of incubation moisture on reptile development and hatchling phenotypes has not been formally reviewed since the 1990s, and an updated synthesis on this topic is timely and needed given the current pace of global change. We conducted a systematic review of literature wherein experimenters manipulated substrate moisture during oviparous reptile development and then reported an effect on phenotypic traits. We used a meta-analytic approach to determine the general effect of substrate moisture on post-hatching phenotypic traits, specifically incubation duration, hatching success, sex ratio and morphology. In our analysis, we also examined potential factors that may moderate the effect of moisture, including taxonomy, incubation temperature and egg size. Our aim was to synthesize the current literature that details the effects of substrate moisture in oviparous reptiles and highlight the importance of this variable in studies of developmental plasticity. As more research is conducted on a greater diversity of reptile species, we may find these results change with the wide range of moisture needs of this diverse taxonomic group (Flatt et al., 2001; Markle et al., 2021). We hope our approach will give a clearer picture of the over-arching trends and encourage further research on the effects of substrate moisture on the developmental plasticity of ectotherms.

MATERIALS AND METHODS

Data collection

Data were obtained from two sources: the Reptile Development (RepDevo) Database (version 1.0.2 downloaded from www.repdevo.com; Noble et al., 2018a,b) and a systematic Web of Science (v.5.35) literature search. The RepDevo Database contains data from 200

laboratory-based manipulative studies about the effects of incubation environment on reptile phenotype published between 1974 and 2016. This database contains nearly 10,000 phenotypes from 155 reptile species (Noble et al., 2018a,b). Importantly, reptile incubation studies sometimes manipulate both temperature and water potential, and water potential of each treatment was encoded within the RepDevo dataset. We also conducted a Web of Science search for studies that were published between 2016 and 2022, using two sets of predefined search terms. The first set reflected those used in Noble et al. (2018a) that focused on studying experimental manipulations of incubation temperature, because these search terms have previously found published experiments that also include moisture manipulations. The second set of search terms focused on incubation substrate moisture treatments, which addressed our question directly. Our two search strings were:

(1) TOPIC:(temperature*) AND TOPIC:(incubat*), along with one of the following individually added as TOPIC:(alligator*; caiman* chelon*; crocodil*; lizard*; reptil*; snake*; sphendon*; squamat*; testudin*; tuatara*; or turtle*); and

(2) TOPIC:(incubat*) AND TOPIC:(water potential*) OR TOPIC:(moisture*) OR TOPIC:(hydric*), along with one of the following individually added as TOPIC:(alligator*; chelon*; crocodile*; lizard*; reptil*; squamat*; snake*; sphendon*; testudin*; tuatara*; or turtle*).

Overall, 435 peer-reviewed articles were found in the literature searches. Titles and abstracts were screened in Covidence, a systematic review management tool (Kellermeyer et al., 2018). Here, we assessed whether articles presented experiments that incubated eggs under two or more moisture conditions in a laboratory setting within 48 h of laying (i.e. our inclusion criterion). This criterion allowed us to identify and focus on studies that manipulated substrate moisture and minimized variation between possible study designs. If articles focused on viviparous squamates they were also excluded, because the moisture conditions that embryos experience cannot be held constant in these species. After screening article titles and abstracts, we were left with 130 articles. These papers were read in their entirety and data were extracted if the study's methods met our inclusion criteria and reported data on at least one or more of the following traits: incubation duration, hatching success, morphology (length or mass) or sex ratio (proportion male). At this stage, papers were excluded for a variety of additional reasons: they did not incubate embryos in the laboratory ($N=47$), they did not report substrate moisture ($N=30$), they incubated eggs within a single moisture treatment ($N=19$), they were duplicates of papers already in the RepDevo Database ($N=8$), they reported trait data outside of our focus ($N=5$), we could not convert reported moisture values to kPa ($N=2$), they did not report error values ($N=2$), we could not get access to the full text ($N=2$), they included confounding factors ($N=2$), they changed moisture concentrations throughout incubation ($N=1$), they experimented on adult reptiles ($N=1$), or they had fluctuating moisture treatments ($N=1$) (Fig. 1). We contacted authors for missing data or additional data if it seemed likely that this information was available but not published and, using this method, received data from three studies. In total, we extracted data from 10 articles from our Web of Science literature search. One additional paper was provided by an author (D.W.), which brought our total to 11. We also applied our inclusion criterion to the articles in the RepDevo database and that separately resulted in 23 articles from which we could extract data. Finally, during the peer-review process, 13 additional papers were screened (Table S2) and three were identified that had extractable data (Packard et al., 1980, 1983; Booth and Yan Yu, 2009).

Thus, in total, we extracted data from 37 articles (Fig. 1) representing a broad range of taxonomic and geographic diversity.

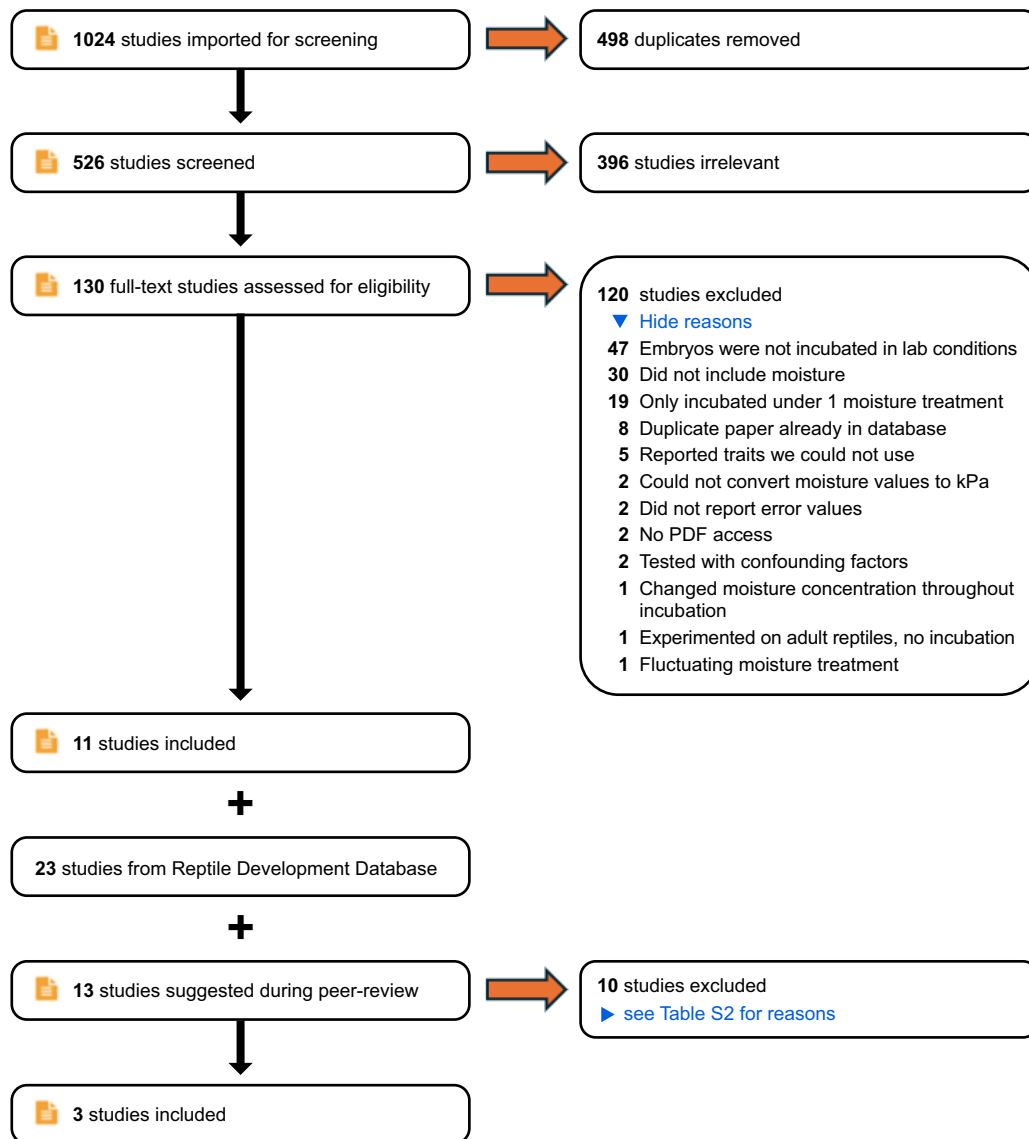


Fig. 1. PRISMA data describing studies excluded at each stage of our systemic literature review. The first box displays the number of studies generated from our search terms, the second indicates the number screened by abstract, and the third box indicates studies that were screened by full text. In total, 11 studies were included after the screening was complete, and these were added to the appropriate papers from the Reptile Development Database ($N=23$) for data extraction. Another 13 studies were suggested for consideration by a peer reviewer, and three of these had data that were extracted and included in our database.

This dataset included 28 different species of squamates ($N=17$) and testudines ($N=11$). Most species were unique to a single study, though some species were included in multiple studies, including the common snapping turtle, *Chelydra serpentina* (5 studies), common keelback, *Tropidonophis mairii* (3 studies), Cuban brown anole, *Anolis sagrei* (2 studies), painted turtle, *Chrysemys picta* (3 studies), eastern short-necked turtle, *Emydura macquarii* (2 studies), Suter's skink, *Oligosoma suteri* (2 studies) and fence lizard, *Sceloporus undulatus* (2 studies). Studies were conducted in the USA ($N=13$), China ($N=9$), Australia ($N=7$), Spain ($N=4$), New Zealand ($N=2$), Canada ($N=1$) and Cuba ($N=1$). Studies also used different incubation substrates. The majority (76%) used vermiculite ($N=28$), others used sand ($N=4$), some studies used both sand and vermiculite ($N=2$), one study used filter paper ($N=1$) and the others did not report the substrate that was used ($N=2$).

Data extraction

For each of the 37 articles in our final dataset, we extracted sample size and trait data from substrate moisture treatments. Paired treatments consisted of two different substrate moisture levels at the same mean temperature. Some studies included treatments with multiple levels of substrate moisture and temperature. For these studies, we extracted all possible pairwise substrate moisture treatment comparisons, but always at the same mean temperature. Some studies had an uneven number of treatments within studies; in these cases, we did not include substrate moisture treatments that could not be paired with a contrasting moisture treatment at the same mean temperature value (Table S1) (Muth, 1980; Gutzke and Paukstis, 1983; Paukstis et al., 1984; Spotila et al., 1994; Alberts et al., 1997; Lin and Ji, 1998; Ji and Braña, 1999; Ji and Du, 2001a,b; Gatto and Reina, 2020; Hall et al., 2022; Wang et al., 2021; Wei et al., 2021).

We calculated the difference in substrate moisture between the paired treatments. Not all studies reported substrate moisture in the same units, so we converted the substrate moisture levels to kPa using the protocol described in Packard et al. (1987). Specifically, we used the water potential curve for water content (g g^{-1} dry mass) and converted its value to water potential (kPa) for sand and vermiculite substrates (Packard et al., 1987). Data on traits were extracted as either proportions (i.e. hatching success and sex ratio) or means with their associated standard errors or standard deviations (i.e. incubation duration, mass and length). All standard errors were converted to standard deviation. When data were not available in tables or text, but presented in figures, we used the 'metaDigitise' R function and package to extract the appropriate data (Pick et al., 2019). In addition to the data on traits, we also included data on the average egg mass of the study species to investigate its influence on the effect of substrate moisture (Table S1). Egg mass was typically reported, but in the few studies where it was not ($N=2$ on *S. undulatus*), we used average egg mass reported in other literature for that species (Ferguson and Talent, 1993).

Effect size calculation

We investigated the effect of substrate moisture on the treatment-level proportions and mean value of quantitative traits. For proportional traits (i.e. proportion hatched and proportion male) we calculated the log odds ratio (lnOR), which gives the relative difference in the odds of an outcome from the wetter treatment compared with the drier treatment (Baker, 2022; Raynal et al., 2022). Incubation duration, length and mass are quantitative traits, so we calculated effect size using the log response ratio (lnRR) that represents the log of the ratio in means between wet and dry treatments (Hedges et al., 1999; Lajeunesse, 2011; Raynal et al., 2022). We made a few modifications to our data to perform these calculations. For example, for control/treatment groups containing a '0' we added '0.1' (Woolf, 1955), so the effect size could be defined. We also converted negative water potential values (kPa) to positive values, so that the absolute differences were used in these calculations; values closer to zero represent substrates with more water available for uptake by eggs. All experimental treatments within our dataset were within negative kPa values, so we had no issues with miscalculation of the absolute differences. Consequently, we refer to substrates with water potentials close to zero as 'wet', and those increasingly further from zero as 'dry'. Effect sizes were calculated so that positive values indicate an increase in a trait in the wet treatment compared with the dry treatment. Effect sizes were calculated using R (version 4.2.1; <https://www.r-project.org/>) using the function 'escalc' in the 'metafor' R package (version 3.8-1; Viechtbauer, 2010).

Meta-analysis

We began by running a multi-level meta-analysis (MLMA) (intercept-only model) that contained three random effects: phylogeny, paper identity (ID) and effect size ID. This MLMA used a residual maximum likelihood (REML) which obtains a mean estimate of the effect size of substrate moisture and estimates the degree of heterogeneity (I^2) using the 'orchard' R package (version 2.0; Nakagawa et al., 2021). I^2 is the percentage of total effect size variance excluding sampling error (also called I^2_{total}). An I^2 of 75% is considered high, 50% medium and 20% low (Higgins et al., 2003), although multi-species analyses such as ours commonly have I^2 values of greater than 90% (Senior et al., 2016).

Our dataset contains multiple species that, in some cases, are repeated across studies and have varying phylogenetic relatedness. We accounted for this nonindependence by including phylogeny as a

random effect (Chamberlain et al., 2012; Noble et al., 2017; Cinar et al., 2022). To do this, we created a phylogenetic tree that contained all the species in the dataset using the Interactive Tree of Life online generator (version 3.0.12; <http://itol.embl.de/>) via the 'rotl' R package (Michonneau et al., 2016). This created a topological tree that we then converted to an ultrametric tree using the 'compute.brlen' function from the 'ape' R package (version 5.6.2), so that our tree would contain branch length data (Paradis and Schliep, 2019). We also accounted for studies containing multiple effect sizes and within-study effect size variance using a unique identifier for each effect and paper as random effects.

Meta-regression

We used meta-regressions to determine whether the effect of substrate moisture is influenced by the following moderators: magnitude of difference in substrate moisture treatments (continuous), incubation temperature (continuous), taxonomic order (categorical: Testudines/Squamata), egg mass (continuous), and the interaction between incubation temperature and the magnitude of difference in substrate moisture treatments. First, temperature was included as it is currently the most well understood factor that affects reptile development, and accounting for its impact could help reveal an additional impact of moisture. Second, taxonomy was included because variation in eggshell structure (Packard, 1991; Debruyne et al., 2023), developmental stage at laying and sex determining mechanism (Valenzuela and Lance, 2004) may differ between taxa, as well as other factors that might be the result of shared ancestry. Third, the inclusion of egg mass accounts for how size may impact the way an egg absorbs and uses moisture for development. Egg mass was extracted from the study papers (reported as means), though as mentioned above (see 'Data extraction' section), some had to be found from other sources (Ferguson and Talent, 1993). Fourth, difference in substrate moisture treatments was calculated by subtracting the wettest from the driest treatment within an effect size pair, because wet treatments had smaller kPa values. Including the difference between treatments accounts for 'nuisance heterogeneity' and allows us to standardise substrate moisture differences across studies (Noble et al., 2022). To compare moderators, we z-transformed egg mass using the 'scale' R function from the 'base' R package (version 4.2.2) and scaled temperature and substrate moisture by 25°C and $[-320]$ kPa, respectively.

Meta-regression models were run in the 'metafor' R package (Viechtbauer, 2010) using maximum likelihood, and included the same random effects as in our MLMA detailed above (phylogeny, paper ID and effect size ID). During model construction, we visually inspected plots to explore whether nonlinear relationships may be present in our data. Overall, these plots suggested linear relationships. It is possible that nonlinear relationships may exist between the traits examined and substrate moisture. Yet, there were insufficient data, particularly at very large water potential differences, to explore the possibility.

To determine which moderators should be included in the best-fitting model, we performed model selection using Akaike's information criterion for small sample sizes (AICc). In doing so, we used the 'dredge' R function from the 'MuMIn' R package (version 1.47.5). For model selection, we only considered models that included both temperature and moisture at the lowest AICc level. ΔAICc values were calculated based on the difference between the focal model and the AIC value of the lowest model containing both temperature and moisture. Moderators within the lowest AICc were used in a final model to estimate the effect of each moderator using a meta-regression with the REML method.

Publication bias

Publication bias was tested for each trait category. To accomplish this, we ran a model using the 'rma.mv' function from the 'metafor' R package (Viechtbauer, 2010) that included all the moderators and random effects in the best-supported model with the addition of a publication year and sampling variance moderator (Nakagawa et al., 2022).

RESULTS

MLMA: The effect of substrate moisture on average phenotype

Substrate moisture had a significant positive effect on body length and mass (Table 1). Hatchlings in wet treatments were longer and heavier than those in dry treatments. Substrate moisture had a weak effect on all other phenotypic traits (hatching success, sex ratio and incubation duration) with confidence intervals overlapping zero for all three effect size measurements, thus reflecting a lack of significance (Table 1). Hatching success ($I^2=50.67\%$) and sex ratio ($I^2=47.49\%$) had lower heterogeneity than body length ($I^2=92.14\%$), mass ($I^2=93.89\%$) and incubation duration ($I^2=92.52\%$) (Table 1).

Meta-regression of moderators for the effect of substrate moisture on average phenotype

Two factors influenced the effect of substrate moisture on hatching success. First, as the moisture difference between treatments became larger, the magnitude of the effect on hatching success increased (Table 1, Fig. 2A). For example, when holding mean temperature at 25°C, a 1 kPa increase between treatments led to a 0.1% increase in the impact of moisture on hatching success. Temperature was also an important modifier; specifically, warmer incubation temperatures diminished the impact of water potential on hatching success (Fig. 2C). For example, when holding the mean water potential difference between moisture treatments at $[-320]$ kPa, a 1°C increase in temperature decreased the chance of hatching success by 3.7%.

The magnitude of the effect on sex ratio was not influenced by the difference between moisture treatments (Fig. 2B) or incubation temperature (Fig. 2D). However, our model for sex ratio indicated statistical support for an interaction between temperature and substrate moisture difference, but its effect was negligible ($\beta=-0.001$) (Table 1). For example, at a mean moisture difference of $[-320]$ kPa and a mean temperature of 25°C, a 1 kPa increase in the difference between treatments and a 1°C increase in temperature decreased the impact of water potential in wet nests compared with dry nests by 0.1% (Fig. 3). In addition, the same difference in moisture treatments led to a significantly greater difference in the proportion of males between wet and dry nests for testudines when compared with squamates (Table 1, Fig. 4). Sex ratio was the only trait affected by the 'Order' moderator.

The moisture differences between treatments influenced the effect of substrate moisture on body mass. As the moisture difference between treatments became larger, the magnitude of the effect on mass increased (Table 1, Fig. 5A). For example, when holding mean temperature at 25°C, a 1 kPa increase in the difference between treatments led to a 0.01% increase in the impact of moisture on mass.

For all other traits, temperature and the difference in moisture between treatments had negligible, non-significant impacts on the effect of moisture (Table 1, Fig. 5). Although it was included in our analysis, egg mass had no significant effect on any trait, nor did it appear in any final models as determined through our model selection process.

Publication bias

Publication year was a significant predictor of lnOR for hatching success ($\beta=0.409$, $P=0.008$, $CI=0.108$, 0.710 , corrected MLMA value=0.1878, $CI=-0.3647$, 0.7403) and lnRR for incubation duration ($\beta=-0.001$, $P=0.042$, $CI=-0.002$, -0.000 , corrected MLMA value=0.0006, $CI=-0.0087$, 0.0098). In contrast, publication year was not a significant predictor of lnOR for

Table 1. Effect of wet (versus dry) soil incubation treatments on the phenotype of reptiles

Meta-regression											
Trait	N_e	N_{st}	N_{sp}	I^2 (%)	MLMA	Order	Moisture difference	Temperature	Moisture difference: Temperature	Egg mass	AIC
Hatching success	202	22	22	50.67	InOR=−0.062 CI=−0.344, 0.220 PI=−1.816, 1.692	−	β =0.001 CI=0.001, 0.001 P <0.001	β =−0.40 CI=−0.076, −0.003 P =0.034	−	−	Loglik=−348.672 AICc=709.8 Δ =0.0
Length	150	23	22	92.14	InRR=0.015 CI=0.002, 0.027 PI=−0.092, 0.122	−	β =0.000 CI=−0.000, 0.000 P =0.930	β =−0.001 CI=−0.002, 0.002 P =0.652	−	−	Loglik=214.854 AICc=−417.1 Δ =0.5
Mass	229	23	23	93.89	InRR=0.030 CI=0.008, 0.052 PI=−0.193, 0.253	−	β <0.001 CI=−0.000, 0.000 P <0.001	β =−0.001 CI=−0.005, 0.003 P =0.618	−	−	Loglik=149.012 AICc=−285.6 Δ =1.9
Sex ratio	80	12	12	47.53	InOR=−0.533 CI=−1.184, 0.118 PI=−2.596, 1.529	β =−1.473 CI=−2.854, −0.091 P =0.040	β =0.001 CI=−0.001, 0.001 P =0.286	β =0.018 CI=−0.111, 0.147 P =0.750	β =−0.001 CI=−0.000, −0.000 P =0.027	−	Loglik=−131.397 AICc=280.8 Δ =1.2
Incubation duration	119	19	20	92.52	InRR<0.000 CI=−0.007, 0.008 PI=−0.059, 0.060	−	β =0.000 CI=−0.000, 0.000 P =0.279	β =0.002 CI=−0.000, 0.003 P =0.049	β =−0.000 CI=−0.000, 0.000 P =0.075	−	Loglik=249.522 AICc=−484.0 Δ =1.09

Parameter estimates are shown with 95% confidence intervals (CI) and 95% prediction intervals (PI) from the multi-level meta-analysis (MLMA) model and the top Akaike's information criterion (AICc) model of the meta-regression for each trait analysis. We also report the heterogeneity percentage (I^2 ; MLMA model). Bold indicates significance; '—' denotes a factor was included in the full model but not in the best-fit AICc-selected or final model. We also include the AICc value for the best-fit model, as well as the Δ that represents the difference in AICc models between the best-fit model and the next-closest model in model selection. N_e represents the number of effect sizes, N_{st} the number of studies, and N_{sp} the number of species for each dataset. Egg mass was z-transformed, and temperature and substrate moisture difference were mean centred to 25°C and $[-320]$ kPa, respectively.

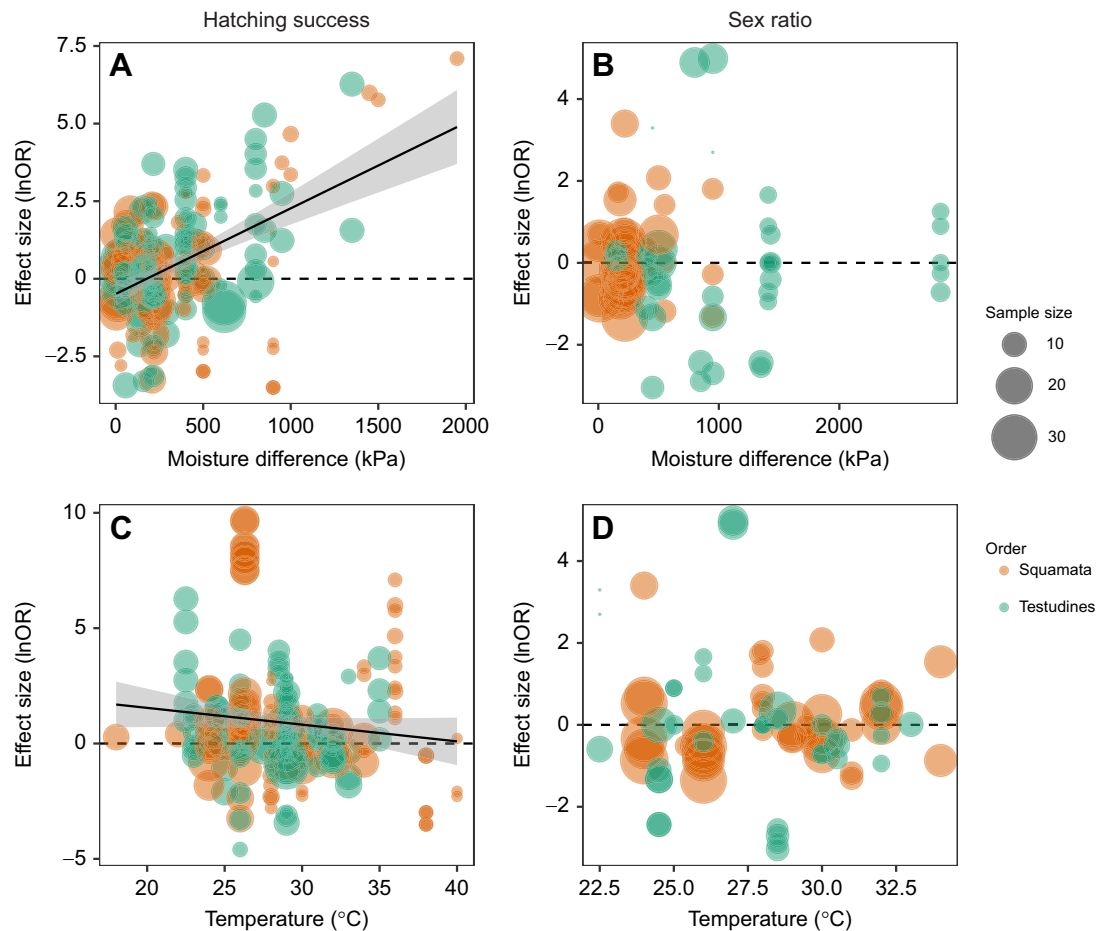


Fig. 2. Effect sizes of mean phenotype in wet versus dry substrate moisture treatments as a function of substrate moisture difference (kPa) and temperature (°C) for hatching survival and sex ratio. We plotted the log odds ratio (lnOR) for hatching success ($N=202$) against (A) difference in substrate moisture ($P<0.001$) and (C) temperature ($P=0.034$). We also plotted the log odds ratio for sex ratio ($N=80$) against (B) difference in substrate moisture and (D) temperature. Effect size estimates above zero indicate that reptiles from the wet treatment had a higher ratio of males to females or higher survival than those from the dry treatment; values below zero show a higher ratio of females to males. Shading on the trendline represents a 95% confidence interval.

sex ratio ($\beta=0.040$, $P=0.138$, $CI=-0.013$, 0.092), lnRR for mass ($\beta<-0.001$, $P=0.66$, $CI=-0.003$, 0.002) or lnRR for length ($\beta<-0.001$, $P=0.944$, $CI=-0.012$, 0.011).

Sampling variance was a significant predictor of effect size variance (Nakagawa et al., 2022) for mass ($\beta=0.030$, $P=0.008$, $CI=0.008$, 0.052), length ($\beta=0.014$, $P=0.015$, $CI=0.003$, 0.026) and sex ratio ($\beta=-0.511$, $P=0.019$, $CI=-0.938$, -0.083). Sampling variance was not a significant predictor of effect size variance for hatching success ($\beta=-0.061$, $P=0.652$, $CI=-0.327$, 0.205) or incubation duration ($\beta=0.001$, $P=0.772$, $CI=-0.006$, 0.008).

DISCUSSION

Our meta-analysis across 28 different species of squamates and turtles from 37 studies showed that substrate moisture impacts reptile embryonic development largely through changes in body size (length and mass). For other phenotypic traits – specifically hatching success, sex ratio and incubation duration – the impact of substrate moisture was affected by certain moderating factors. Specifically, the effect of moisture on sex ratio differed between reptile orders, the effect of moisture on hatching success was influenced by temperature, and the effect of moisture of hatching success and hatchling mass was moderated by the magnitude of the difference between moisture treatments. The interaction between temperature and the difference in the substrate moisture's effect on

sex ratio is interesting because substrate moisture has rarely been shown to affect sex ratio, and temperature is the predominant factor influencing sex determination in many reptiles (Valenzuela and Lance, 2004). These results support the idea that an interaction between temperature and moisture, possibly through evaporative cooling and water vapor exchange, could be a significant moderator of sex ratio determination (Packard, 1991; Packard and DeMarco, 1991). Importantly, effect sizes were generally small across our study, and publication bias was present for some traits. Specifically, the positive effect of publication year on hatching success may indicate a selective publication bias in recent literature; in contrast, for incubation duration, smaller effect sizes are reported in more recent studies, which may indicate increased reporting of this trait in recent literature. Overall, research on the effects of moisture on reptile development has not yet received a major quantitative synthesis; thus, our study provides a major advancement in understanding this important topic.

Morphological traits, specifically length and mass, were positively affected by substrate moisture – wetter treatments produced longer and heavier hatchlings. The results found in this study agree with previous work in reptiles, including turtles (Gatto and Reina, 2022) and lizards (Phillips et al., 1990; Warner and Andrews, 2002b). This trend may be explained by the fact that increased water availability during incubation can result in reptiles converting more yolk mass

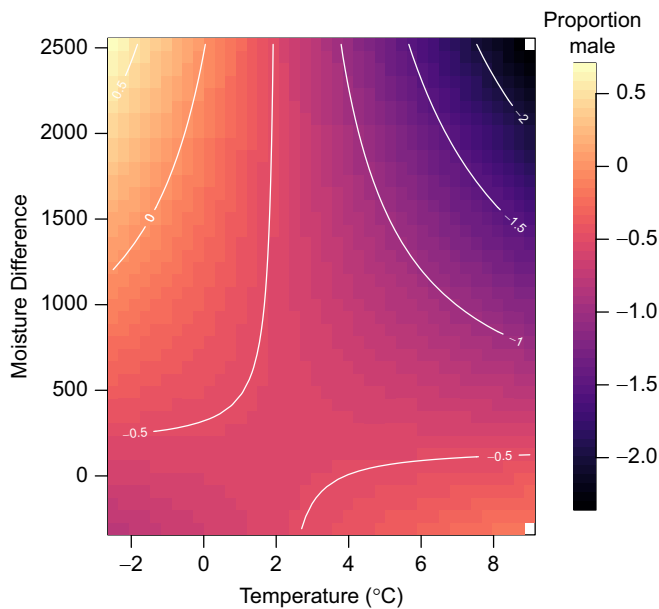


Fig. 3. Contour plot depicting the impact of the interaction between temperature and moisture difference on sex ratio (proportion male). Moisture difference and temperature values are mean centred based on a water potential of $[-320]$ kPa and temperature of 25°C . As the key shows, colours lighten as the proportion of male offspring increases and darken as the proportion of male offspring decreases.

into body mass, thus producing larger hatchlings (Packard et al., 1988; Hewavisenth and Parmenter, 2001; but see Miller and Packard, 1992). This may be especially true for many squamates and some turtles that typically lay soft-shell eggs, as this eggshell type has a higher permeability of water through its surface, which can

increase water uptake and, in turn, facilitate the catabolism of yolk to tissue, resulting in larger offspring (Belinsky et al., 2004; Booth, 2004; Deeming and Unwin, 2004). Indeed, eggs of the red-eared slider (*Trachemys scripta*) incubated in extremely dry environments (i.e. -1000 kPa or the equivalent to a drought) produced smaller hatchlings the longer they were exposed to these harsh conditions (Delmas et al., 2008). Rigid-shell eggs, like those of some turtles and crocodilians, are laid by their mothers with enough internal water to complete embryonic development; in contrast, soft-shell eggs, such as those of most squamates, require more water than the amount they are laid with and must acquire it from the surrounding nest environment (Vleck, 1991; Belinsky et al., 2004; Booth, 2002, 2004; Booth and Yan Yu, 2009). Unlike rigid-shell eggs, soft-shell eggs have a lower resistance to water exchange across the eggshell, which is typically quantified as water conductance (Vleck, 1991; Belinsky et al., 2004; Booth, 2002, 2004; Booth and Yan Yu, 2009). Thus, eggshell type should be considered in future research on this subject.

Substrate moisture affected sex ratio for testudines but not squamates. The occurrence of temperature sex determination (TSD) and genetic sex determination (GSD) varies among reptilian taxa. Within the species studied in papers selected for our meta-analysis, there was a predominance of TSD in testudines (89%) compared with squamates ($\sim 5\%$), and this reflects patterns reported in the literature (Valenzuela and Lance, 2004; Michonneau et al., 2016). This disparity is likely responsible for our finding that sex ratios are influenced by developmental moisture more in turtles than in squamates. Importantly, this impact occurs in the laboratory even after experimentally controlling for temperature, indicating that the masculinising effect of higher substrate moisture acts independently of other environmental factors. The turtles included in our dataset are TSD 1A species, wherein more males are produced at cooler temperatures and more females at hotter temperatures (Valenzuela

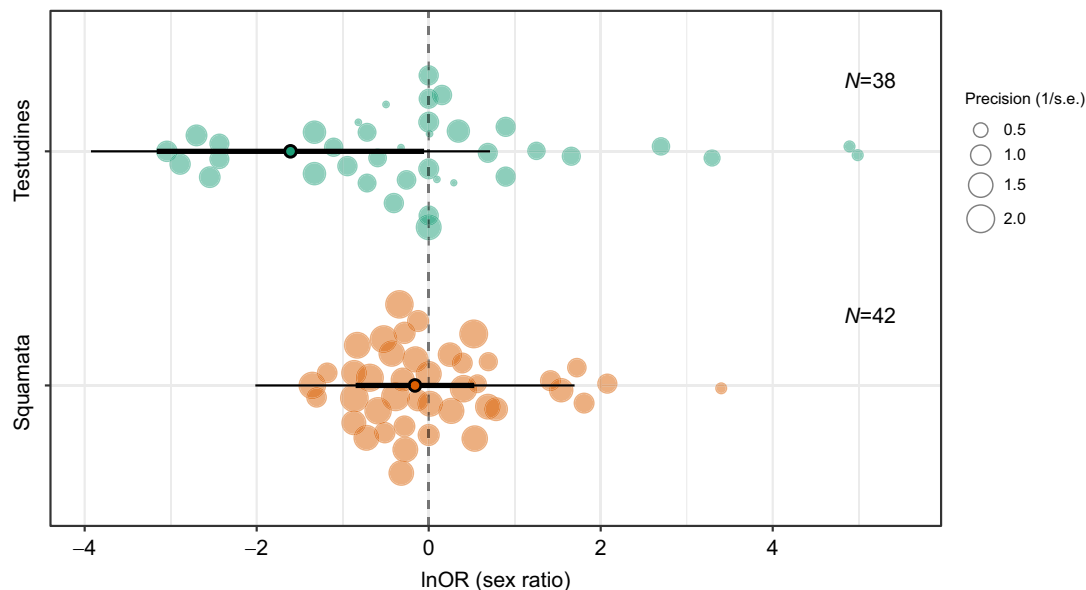


Fig. 4. Orchard plot describing the differing impact of taxonomic order on the effect of substrate moisture on sex ratio. Depicted are marginalised means whose estimation is conditioned on the other factors in the final model (Table 1). These are represented by the small, coloured circles outlined in black. The wide horizontal line signifies a 95% CI, and if this line does not intersect a log odds ratio (lnOR) of 0, then the result is significant. We also plot the effect sizes (lnOR), which are sized by their precision (1/s.e.) and are predicted based on the other factors included in the final model. Effect sizes less than 0 indicate a higher ratio of female hatchlings in wet treatments, whereas values greater than 0 indicate a higher ratio of male hatchlings. The number of effect sizes for Testudines (green) and Squamata (orange) are represented by N . This visualisation was generated using the R package 'orchard' (version 2.0; Nakagawa et al., 2023).

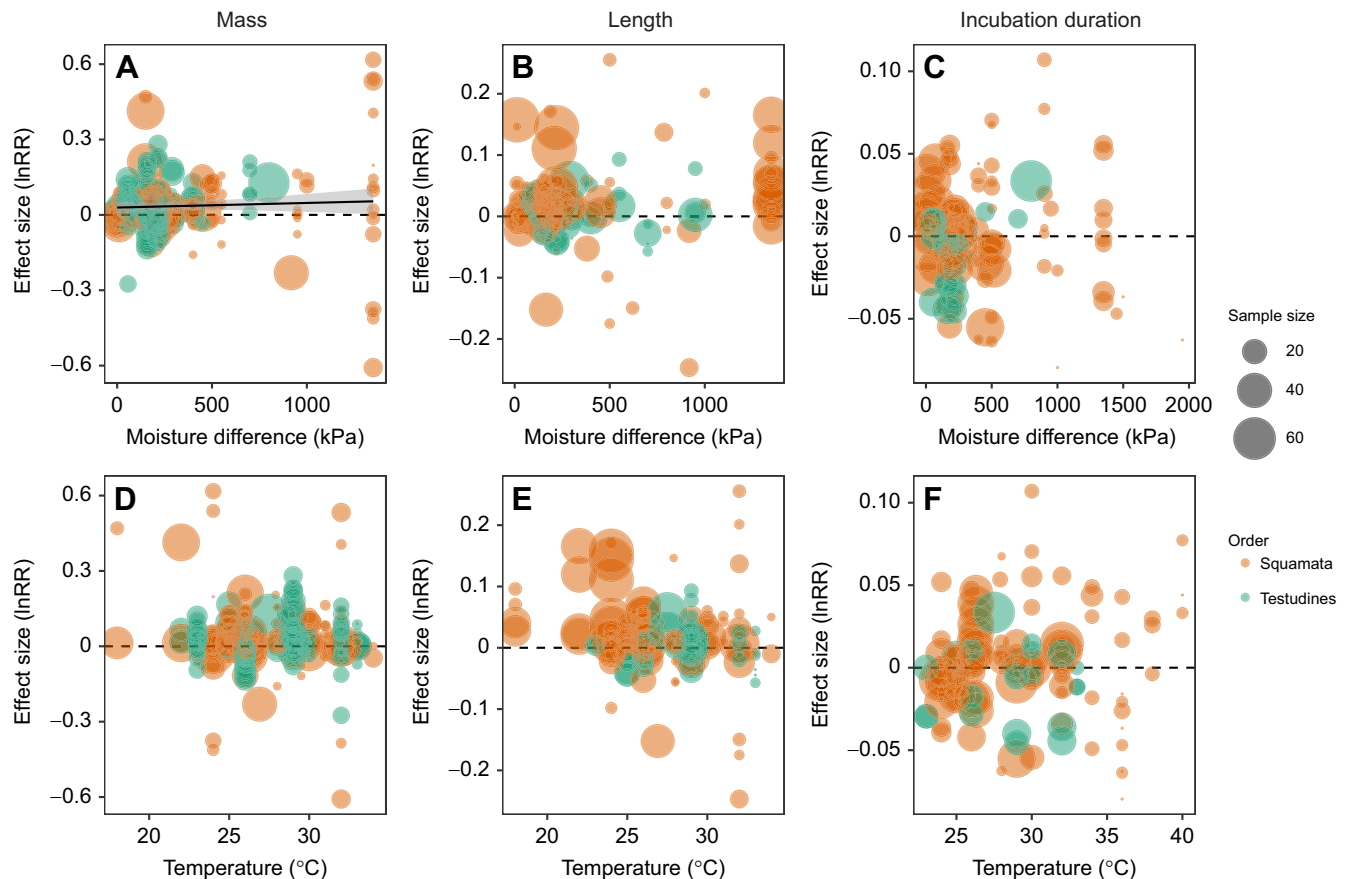


Fig. 5. Effect sizes of mean phenotype as a function of substrate moisture difference (kPa) and temperature (°C) for mass, length and incubation duration. We plotted the log response ratio (lnRR) for mass ($N=229$) versus (A) difference in substrate moisture ($P<0.001$) and (D) temperature. We also plotted the log odds ratio (lnRR) for length ($N=150$) against (B) difference in substrate moisture and (E) temperature. Finally, we plotted the log response ratio (lnRR) for incubation duration ($N=119$) versus (C) difference in substrate moisture and (F) temperature. Values above zero reflect that reptiles from the wet treatment had a longer incubation duration or were heavier or longer than those from the dry treatment.

and Lance, 2004). Research has begun to elucidate how temperature epigenetically mediates this pattern of TSD (Weber et al., 2020; Liu et al., 2022), but the effect of other environmental variables has yet to be considered. Our meta-analysis highlights the importance of also investigating the role moisture has in TSD of oviparous reptiles. In addition, a more focused experimental approach would be useful to advance our knowledge of the impacts of substrate moisture on sex ratio among TSD and GSD species (LeBlanc and Wibbels, 2009; Lolavar and Wyneken, 2017). Indeed, moisture deprivation to pregnant females in viviparous squamates, which are largely GSD species, has been shown to affect sex ratios of offspring (Dupoué et al., 2019). It has previously been suggested that studies that isolate the effects of temperature, substrate moisture and substrate type could empirically tease apart the effects of these environmental factors on sex ratio, as well as other phenotypic traits (Mitchell and Janzen, 2019). This approach, paired with careful selection of model study species (e.g. those with a matching pattern of TSD), may allow an informed comparison of environment effects on sex ratio and would be a fruitful future avenue of research.

Unexpectedly, hatching success was not consistently affected by substrate moisture across studies. Indeed, some studies have also found that hatching success is not significantly affected by moisture treatments within a range that is sublethal (*S. undulatus*; Warner and Andrews, 2002b). Overall, in the literature, it appears that moisture levels that are ideal for hatching success vary greatly across taxa, which likely reflect differences in their habitats. For example,

freshwater turtles (*Emydoidea blandingii*, *Clemmys guttata* and *C. picta*) living in areas with rocky, exposed outcrops seek warm, wet crevice microhabitats (Markle et al., 2021) and eastern three-lined skinks (*Bassiana duperreyi*) in cold climates dig shallow (~3 cm) nests to optimize hatching success (Flatt et al., 2001). So, owing to this variation, a relationship between substrate moisture and hatching success may not be generalizable across taxa. Interestingly, we found that the effect of substrate moisture on hatching success was significantly moderated by temperature. Specifically, the effect of substrate moisture on hatching success was lower at higher temperatures. If the effect of temperature is considered by itself, higher incubation temperatures are known to decrease hatching success (Noble et al., 2018a), though the same is true for lower temperatures because this relationship has a parabolic effect. But how temperature may interact with substrate moisture is not well understood (Ackerman, 1991; Packard, 1991). Conceivably, warmer temperatures might rapidly desiccate nests due to an increased rate of evaporation. This idea is supported by the fact that temperature appears to override impacts of moisture during incubation (Muth, 1980).

Effects of substrate moisture on hatching success and mass were also enhanced by an increase in the magnitude of difference in substrate moisture between treatments. This finding is comparable to other studies showing that very high or very low moisture concentrations can have a greater effect on development (Gatto and Reina, 2022). Most of the studies we used in this meta-analysis did not incubate eggs at the extremes of moisture, as this would not be

ethical nor an ideal design for many studies. This limited number of 'extreme' treatments may hinder our understanding of reaction norms associated with incubation moisture and limit our ability to predict how embryos will tolerate more severe environmental variation. Though, in studies that have used extreme moisture conditions, eggs experienced 100% mortality (Reedy et al., 2013; Warner et al., 2023). Similarly, a study that incubated common chameleon (*Chamaeleo chamaeleon*) eggs under high hydric stress (i.e. a period of 110 days between -1650 and -1950 kPa) and dry incubation conditions (i.e. 100 days at -650 kPa) observed increased hatchling mortality, particularly when eggs were not aggregated (Marco and Díaz-Paniagua, 2008). In general, as demonstrated by these examples, reptile embryos incubated at moisture extremes generally desiccate or drown (Warner and Andrews, 2002a; Markle et al., 2021; Warner et al., 2023). We hope this meta-analysis encourages other studies to cautiously explore the biological limits of substrate moisture to capture the phenomenon of developmental plasticity across a greater range of moisture treatments. This may allow a future meta-analysis to test nonlinear relationships between substrate moisture and hatchling phenotypes in oviparous reptiles (Noble et al., 2018a), which may reveal additional complexities that cannot yet be assessed. Another interesting question that previous research on this topic raises (Marco and Díaz-Paniagua, 2008) is whether there may be interactive effects of egg aggregation and moisture conditions. This is a line of research that may add to the potential benefits of reptile sociality in oviparous species (Marco et al., 2004; Warner and Chapman, 2011). It also may be a particularly important factor for reptiles with large clutch sizes (e.g. sea turtles; Smith et al., 2021), as water exchange may vary based on an egg's position within the nest cavity. Overall, assessment of potential nonlinear effects of soil moisture on reptile development is needed, as it is environmental extremes that are predicted to increase under climate change scenarios.

Despite species-specific examples of high substrate moisture delaying hatching (Gatto and Reina, 2022), we found no generalizable impact of substrate moisture on incubation duration nor moderation by other factors like temperature or taxonomy. This is somewhat surprising given the exceptionally large influence of temperature (and temperature fluctuations) on incubation duration (Noble et al., 2018a; Raynal et al., 2022), suggesting that developmental rate is specifically sensitive to temperature and not moisture (but see Ackerman, 1991; Packard, 1991; Packard and DeMarco, 1991; Vleck, 1991). Across studies, the effect of substrate moisture on incubation duration often varies among populations or species. For example, closely related *Phrynocephalus* species can differ in incubation duration; with mean overall durations being longer for *P. axillaries* and *P. grumgrzimailoi* (Wang et al., 2014). Moreover, incubation duration can differ among distinct populations across different latitudes within the same species (Du et al., 2010). Our findings, as well as other research, suggest that phylogenetics and temperature, rather than substrate moisture, have the most influence on incubation duration of oviparous reptiles.

There were factors we were not able to include in this study; most notably, eggshell structure. The diversity of taxa included in our extracted data did not capture enough variation to test for an effect; the only taxonomic group that exhibited variation in eggshell structure was Testudines. Thus, future research could target taxonomic groups with variation in their eggshell structure, such as turtles (Ackerman, 1991) and geckos (Choi et al., 2018). Additionally, future research could capture the variation observed in eggshell structure by moving away from the traditional classification of eggs as either rigid (crocodiles, some turtles and

most geckos), pliable (turtles) or flexible, parchment-type eggs (most squamates and some turtles) (Packard, 1991). Instead, water absorption rates or other measures of eggshell structure that are continuous would enable greater variation to be captured across species (Debruyne et al., 2023). Future studies and meta-analyses should further explore the effect of eggshell structure on mediating environmental factors and the developmental plasticity of oviparous reptiles.

Overall, our study is the first quantitative meta-analysis exploring the effects of substrate moisture on phenotypes of oviparous reptiles. We set out to analyze and interpret general phenotypic trends that stem from moisture in the nest environment. We found that substrate moisture had a small, but significant effect on hatchling length and mass with wetter treatments producing longer and heavier hatchlings. Two other phenotypic traits were investigated – hatching success and sex ratio – but were only affected by substrate moisture levels after accounting for other factors, such as taxonomy, temperature and difference in substrate moisture. The difference in substrate moisture was also a significant modifier of the effect of substrate moisture on body mass. Possibly, the effect of substrate moisture on phenotype shows nonlinear trends that we were unable to account for owing to the lack of extreme hydric treatments and the smaller number of studies available on this understudied topic. Our study highlights the importance of further research investigating developmental plasticity with respect to substrate moisture.

In conclusion, our research serves as a foundation and launching point for other studies investigating the impact of substrate moisture on developmental plasticity. In this era of rapid, global environmental change (e.g. through urbanization, climate change, etc.), understanding the effects of all environmental conditions, including those less studied, such as moisture, will be critical for predicting how sensitive taxa will respond to these changes. Water is required by all organisms, and the moisture requirements of developing embryos are often overlooked. Yet, successful embryonic development has strong fitness consequences at individual and population levels. Focused studies on the effects of moisture at these early life stages will be critical in understanding how populations will respond to natural and anthropogenically induced changes in the hydric environment.

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Competing interests

The authors declare no competing or financial interests.

Author contributions

Conceptualization: C.B., D.W.A.N., L.E.S., D.A.W., J.L.R.; Data curation: C.B., J.L.R.; Formal analysis: C.B., R.S.R., D.W.A.N., J.L.R.; Funding acquisition: J.L.R.; Methodology: C.B., R.S.R., D.W.A.N., L.E.S., J.E.P., J.L.R.; Project administration: J.L.R.; Supervision: J.L.R.; Visualization: C.B., J.L.R.; Writing – original draft: C.B.; Writing – review & editing: C.B., R.S.R., D.W.A.N., L.E.S., D.A.W., J.E.P., J.L.R.

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Data availability

Data and code are available from an Open Science Framework repository: <https://osf.io/3ptcz/>.

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